DR4: Repairing the Instrument Rather Than the Criterion — and Discovering the Criterion Went Coarse

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair Nomination — DR4 **Status:** H1‴ GO, H2‴ GO, H3‴ GO, H4‴ GO — overall **GO**, with a metric defect reported **Date:** 2026-07-25

Abstract

DR3 passed three of four gates and failed the fourth on a threshold that was unsatisfiable by construction. speedup = expected_random / verifications with verifications ≥ 1, so speedup is bounded above by expected_random, which depends only on the toy's base rate. DR3's costly toy had a 14% base rate, capping speedup at 7× against a 10× gate.

DR4 repairs the instrument and carries the threshold over unchanged. The costly toy's parent budget is tightened from 8 to 1 against commitments of 64, 4 and 2, so that no proper subset of the three costly propositions suffices — all three must be released together. The base rate falls from 191/1350 to 1/1350, matching the restrictive toy, and expected_random rises to 675.5.

All four gates pass. Independence holds (517 candidates with cost_relief > 0 and weakness_gain == 0). Complementarity holds — weakness wins the restrictive toy and is exactly silent on the costly one (tie_fraction = 1.00); cost wins the costly toy and is exactly silent on the restrictive one. All three combiners reach one verification on both toys. Best speedup is 675.5× on both, clearing 10×.

But the GO comes with a defect I am reporting rather than burying. Fixing the ceiling made the criterion coarse. With exactly one load-bearing candidate per toy, speedup is now binary: a nominator that ranks the answer first scores exactly expected_random, and one that does not scores a random draw. There is nothing in between. The 10× threshold no longer discriminates — it has become a restatement of "hit@1." DR4's H4‴ is therefore a *pass*, but a weaker test than DR3's H4″ was intended to be. Section 5 states what a successor must do.

Per the preregistration, a clean sweep opens the date-cut retrodiction. It is open — with the qualification above attached to it.

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1. What was actually wrong with DR3

DR3's H4″ was not a hard test that the nominators failed. It was an arithmetically impossible one:

```
speedup_vs_random = expected_random / verifications_to_first_hit
verifications_to_first_hit ≥ 1
⇒ speedup_vs_random ≤ expected_random
```

and expected_random = (n+1)/(L+1) for n candidates and L load-bearing ones — a property of the toy, not of any nominator. DR3's costly toy had L = 191 of n = 1350, giving expected_random = 7.0. No ordering procedure, however good, could have exceeded 7×.

The nominators had in fact already reached verifications = 1 on both toys — the theoretical optimum. The NO_GO measured my calibration.

DR3's paper named the correct repair: lower the costly toy's base rate, not the threshold. Two repairs were available and they are not equivalent.

The cosmetic repair would be to lower the threshold to 5×. That fits the data by moving the goalpost, and the program has an explicit rule against it.

The structural repair is to change the toy so that fewer deletions qualify. DR4 takes this one. The threshold is untouched.

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2. The recalibrated toy

DR3's costly toy had a single 64-unit commitment against a budget of 8. Any deletion containing sequential_schedule met the budget, so 191 of 1350 candidates qualified.

DR4's CT4 has three costly propositions —

```
sequential_schedule     64.0
checkpoint_every_step    4.0
sync_barrier             2.0
```

— against a budget of **1.0**. Dropping the two largest still leaves 2 > 1. No proper subset suffices; the unique load-bearing deletion is the full triple.

Each of the three remains **vacuous as a predicate**: every hypothesis satisfies it, so deleting any of them leaves the extension identical and weakness_gain == 0 exactly. Only the commitment changes. This is the property DR2 proved unreachable when cost was a minimum over the extension, and it survives the recalibration intact.

The change is more than an arithmetic convenience. It sharpens the analogy the toy is meant to carry. The real move in a representational repair is rarely "drop one thing" — it is drop the constraint *and* discharge whatever the constraint was silently providing. CT4 now requires releasing an entangled set, mirroring the entangled facet triple the restrictive toy has had since DR2.

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3. Calibration, measured before the gates were written

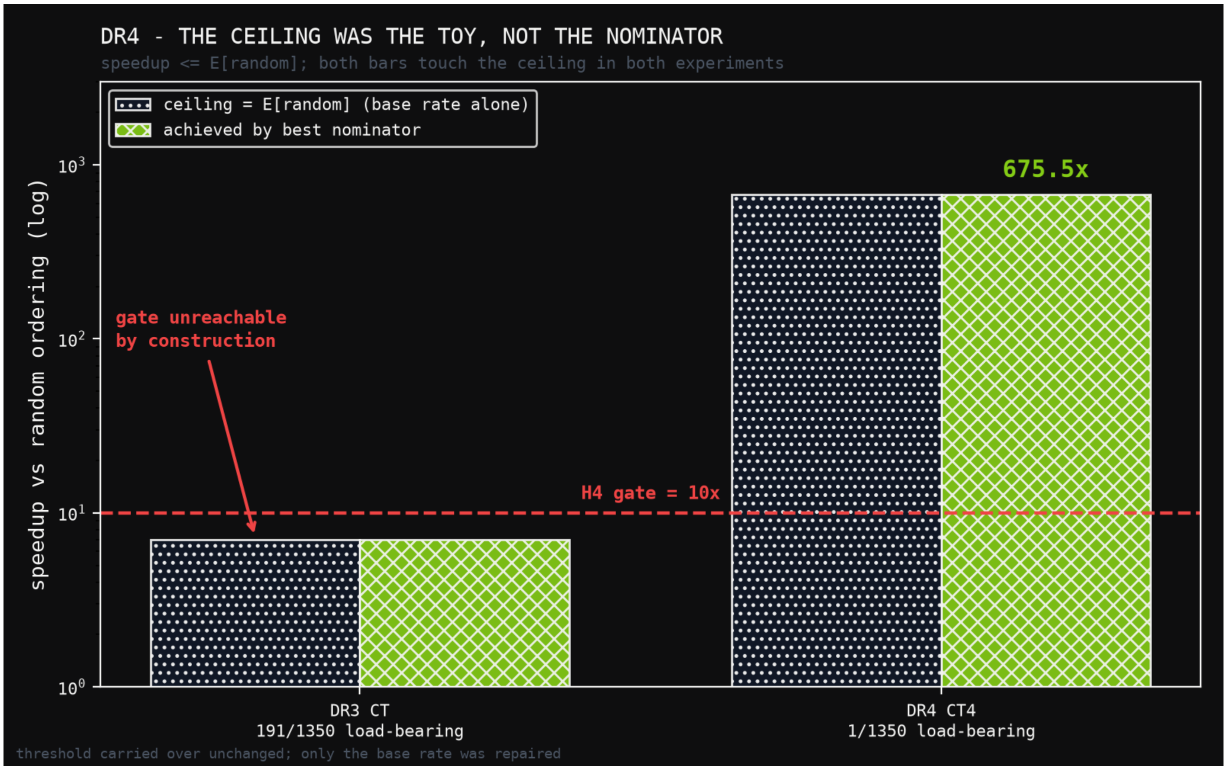
This is the step DR3 skipped, and it is the reason DR4 exists.

toy	candidates	load-bearing	base rate	E[random]	10× needs
RK — restrictive_kinematics	1350	1	0.07%	675.5	≤ 67 verifications
CT4 — calibrated_costly_transduction	1350	1	0.07%	675.5	≤ 67 verifications

Independence, also measured before freezing:

toy	cost>0, weakness=0	weakness>0, cost=0	both>0
RK	0	1	0
CT4	**517**	0	0

DR4_PREREGISTRATION.md was written after this table and before the scored run. That ordering — calibrate, then freeze gates, then run — is what DR3 inverted.



Ceiling

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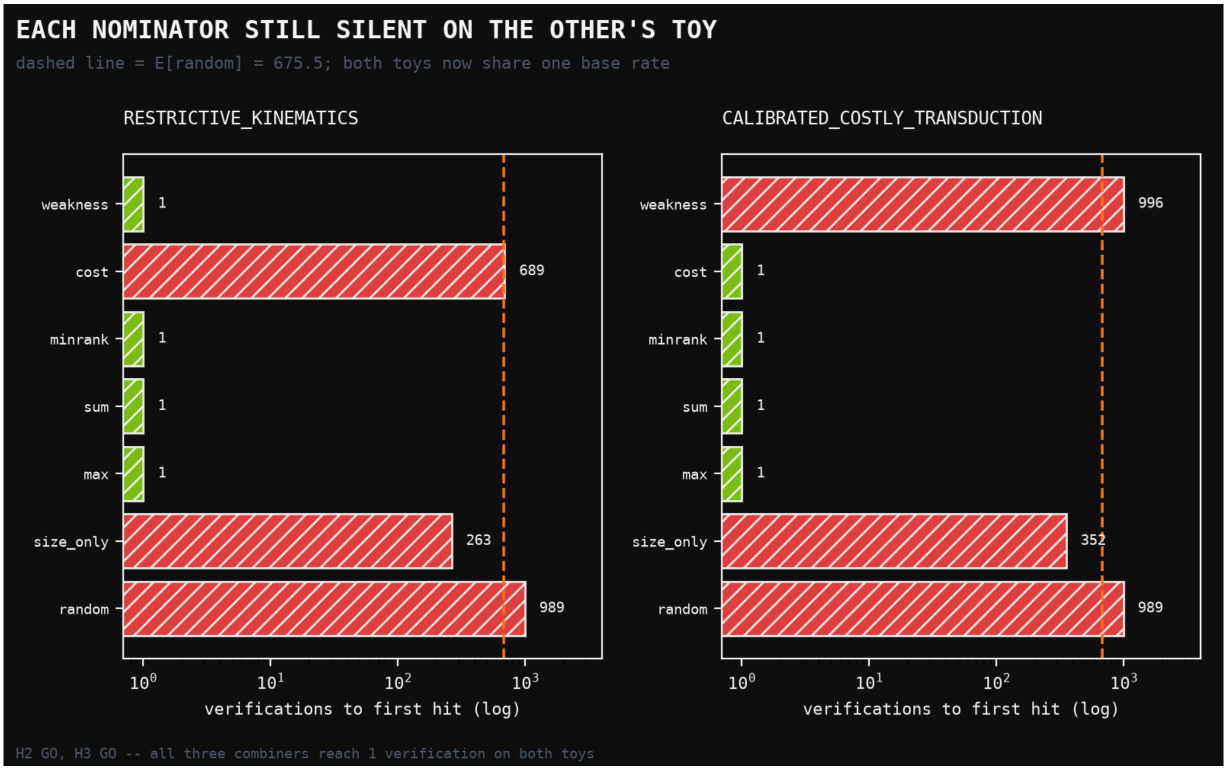
4. Results

nominator	RK verifications	RK speedup	CT4 verifications	CT4 speedup
weakness	**1**	675.5×	996 *(silent)*	0.7×
cost	689 *(silent)*	1.0×	**1**	675.5×
max_disjunction	**1**	675.5×	**1**	675.5×
sum_disjunction	**1**	675.5×	**1**	675.5×
minrank_disjunction	**1**	675.5×	**1**	675.5×
size_only	263	2.6×	352	1.9×
random	989	0.7×	989	0.7×

(silent) marks tie_fraction = 1.00 — the nominator assigns every candidate the same score and contributes no information. Its verification count is then a seeded shuffle draw, not a measurement: weakness's 996 on CT4 should be read as "no signal," not "worse than random."



Complementarity

H1‴ — independence. GO. 517 candidates on CT4 have cost_relief > 0 with weakness_gain == 0. DR2's empty cell stays populated.

H2‴ — complementarity. GO. Best single nominator is weakness on RK and cost on CT4. Each is exactly silent on the other's toy.

H3‴ — combiner. GO. All three combiners reach one verification on both toys, matching the better single nominator everywhere.

H4‴ — speedup. GO. 675.5× on both, against the 10× threshold carried over unchanged from DR3.

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5. The defect in the repaired gate

The GO is real, but it is worth less than it looks, and the reason is a direct consequence of the fix.

With L = 1 load-bearing candidate, a nominator either ranks it first — scoring verifications = 1 and therefore exactly expected_random = 675.5× — or it does not, in which case its position is essentially arbitrary. Every non-degenerate outcome is one of two values. The 10× threshold is cleared by an enormous margin or missed entirely; there is no regime in which it discriminates between a good nominator and a mediocre one.

So DR4 traded an *unsatisfiable* gate for a *coarse* one. H4‴ is now operationally identical to "does the nominator achieve hit@1," and the numeric threshold is decorative. That is a better failure than DR3's — a coarse test that passes is more informative than an impossible test that fails — but it is not the graded speedup measurement the gate was written to be.

The correct successor design is a base rate that is low but not degenerate: roughly 5–20 load-bearing candidates out of 1350, giving expected_random between 65 and 225. Then verifications_to_first_hit can take intermediate values, speedup becomes graded, and a 10× threshold separates nominators that rank well from nominators that rank the answer somewhere in the top fifth. DR4's toys cannot produce that, because the entangled-set construction admits exactly one qualifying deletion by design.

I did not anticipate this when specifying DR4. Calibrating the base rate correctly fixed the ceiling and, in the same move, collapsed the metric's resolution. Recording it here is the same discipline that produced DR4 from DR3's failure.

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6. What DR4 establishes

On toy systems with authored propositions, authored costs, a fixed vocabulary , and an exhaustive oracle over |D| ≤ 3:

weakness gain — 517 candidates exhibit the case DR2 proved impossible.

one toy and exactly silent on the other.

single nominator on both toys.

1350 on both toys.

- Cost defined off the extension yields a signal genuinely independent of
- The two signals are complementary: each is the sole informative nominator on
- Disjunctive combiners exploit both without a tuned weight, matching the best
- Execution-free nomination locates the load-bearing deletion at rank 1 out of

What it does not establish, stated as plainly as the above:

and is findable; cost on CT4 is close to a direct readout of the planted commitment structure. That is why a GO *opens* the retrodiction rather than concluding anything.

framework. Every deletion here is over a fixed proposition set.

- Nothing about real material. Both toys are authored so that the answer exists
- Nothing about vocabulary extension — the known ceiling of the whole
- Nothing graded about nomination quality, for the reason in §5.

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7. What opens

Per DR4_PREREGISTRATION.md §5, a clean sweep of all four gates opens the **date-cut retrodiction**: run the nominators over a real pre-1905 corpus with a hard date cut, and ask whether the load-bearing deletion they nominate is the one the historical record actually made. That has been closed since DR1 and is now open.

It should be entered with §5's qualification attached. The retrodiction will have a naturally occurring base rate, not an authored one, so the speedup metric will be graded there whether or not the toys could produce it. The coarseness is a property of DR4's instruments, not of the measure.

Appendix: reproduction

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uv run --no-sync python -m experiments.deletion_repair.run_dr4
```

Local CPU, seconds. Verdict written to experiments/deletion_repair/results/dr4_verdict.json. Gates frozen in DR4_PREREGISTRATION.md, written before the scored run; calibration in §3 measured before the gates. Ties broken by seeded shuffle (seed = 20260724), never by name — the defect DR1 caught in itself.

Figures: papers/deletion_repair_dr4/figures/build_figures.py.